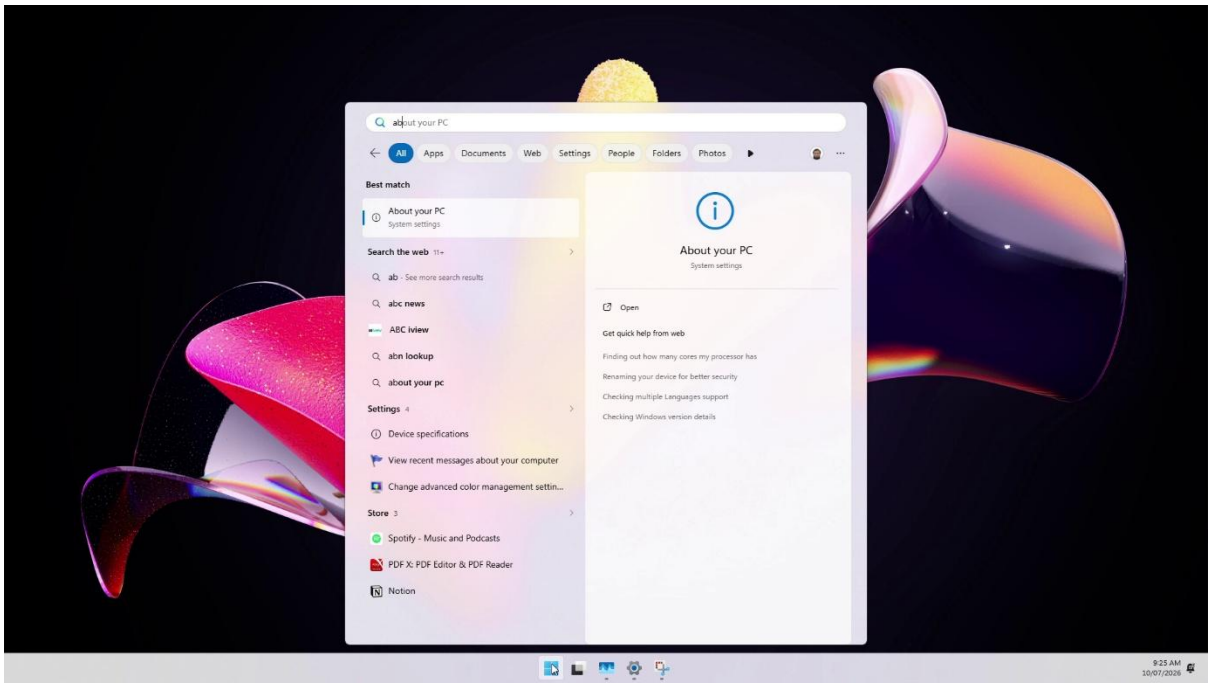
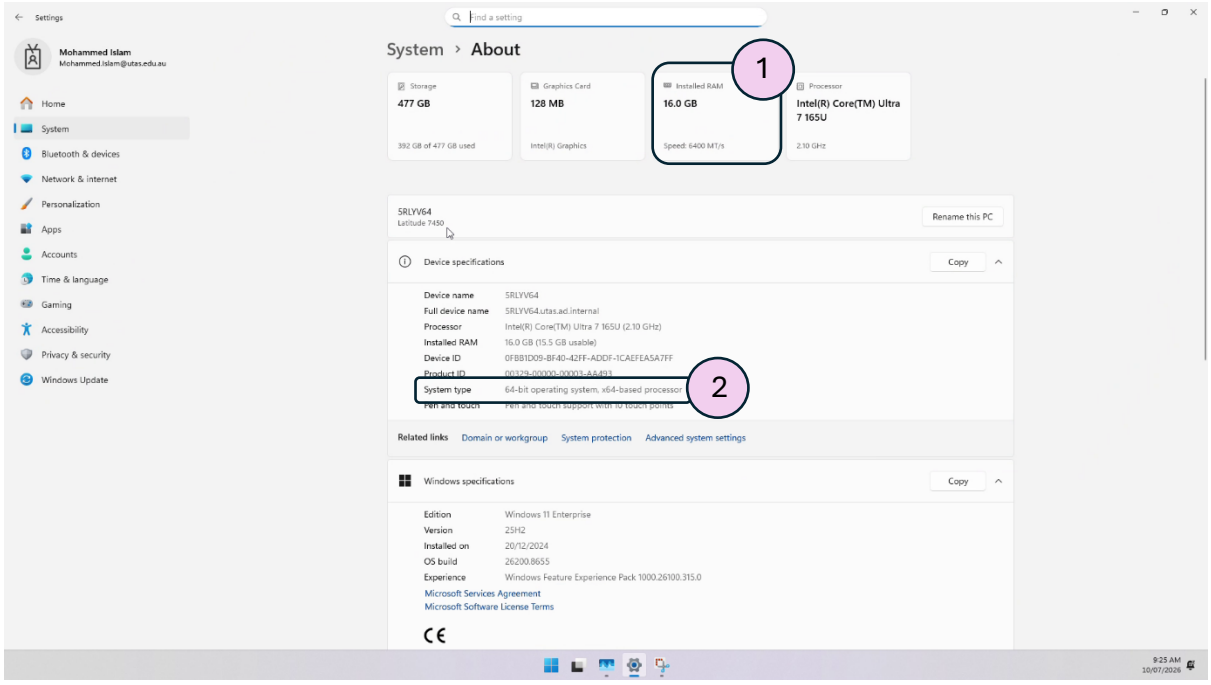
Check Prerequisites

Before installing Docker, verify:

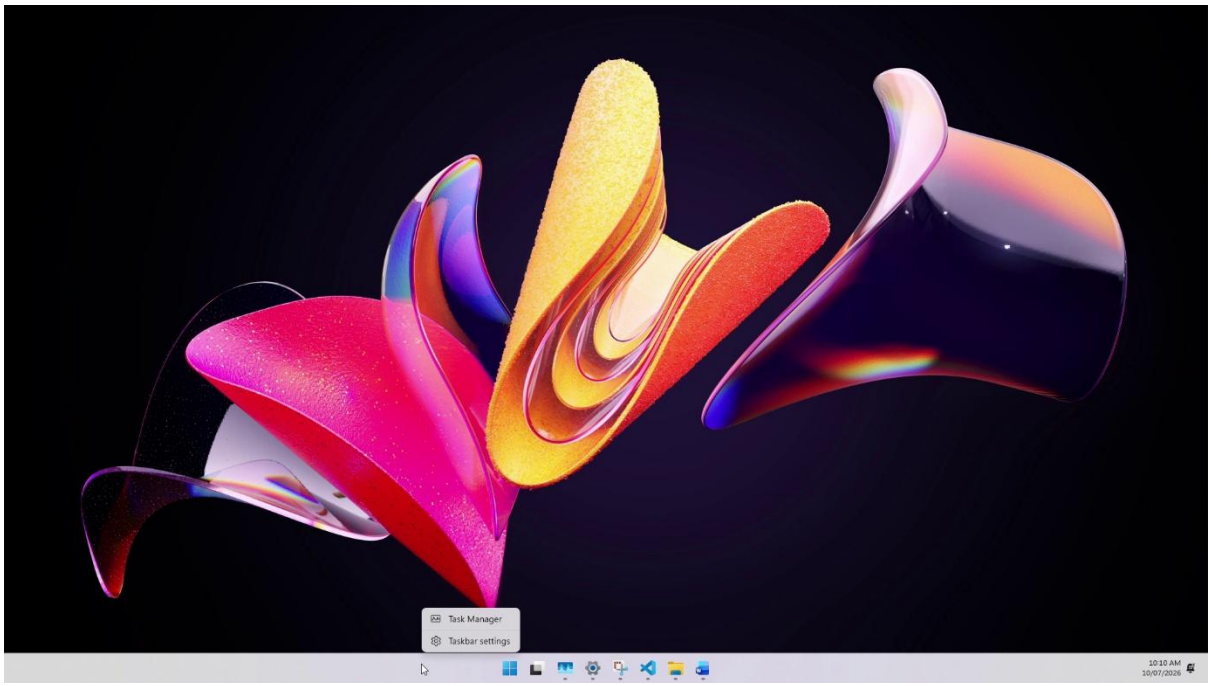
- At least 4 GB RAM (8 GB recommended)
- 64-bit operating system
- Hardware virtualization enabled in BIOS/UEFI

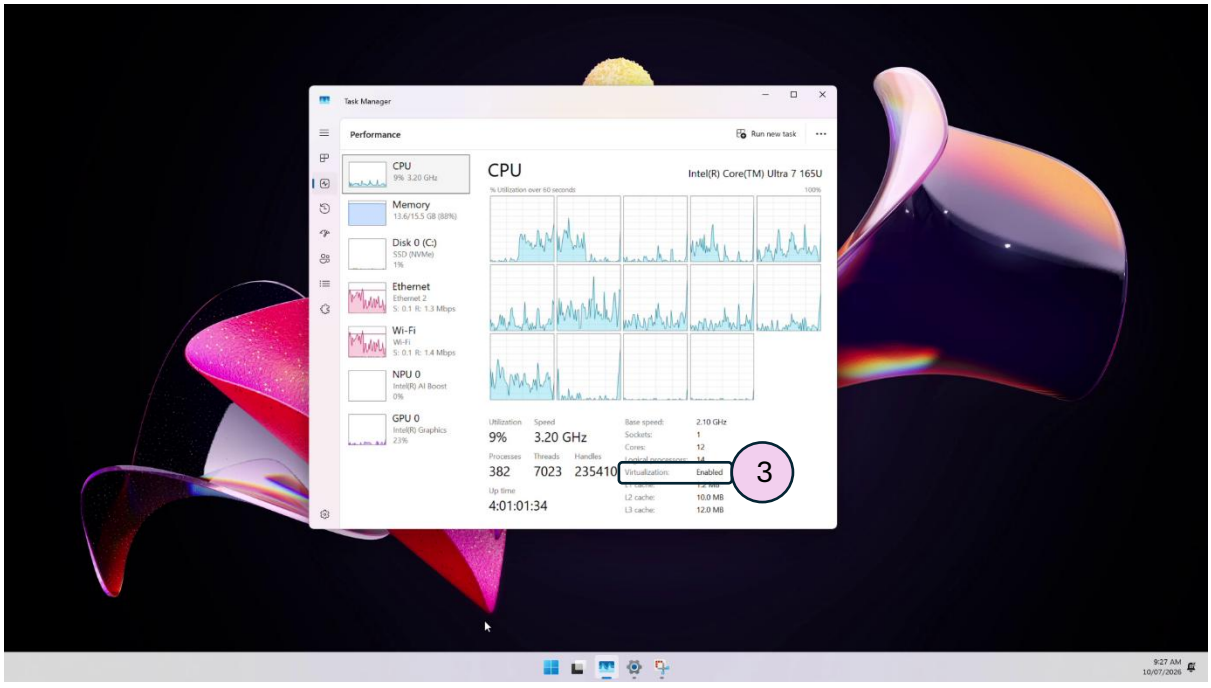
Click on the “Start” button and type “About your PC.”





Open the task manager and check if the “Virtualization” is enabled.

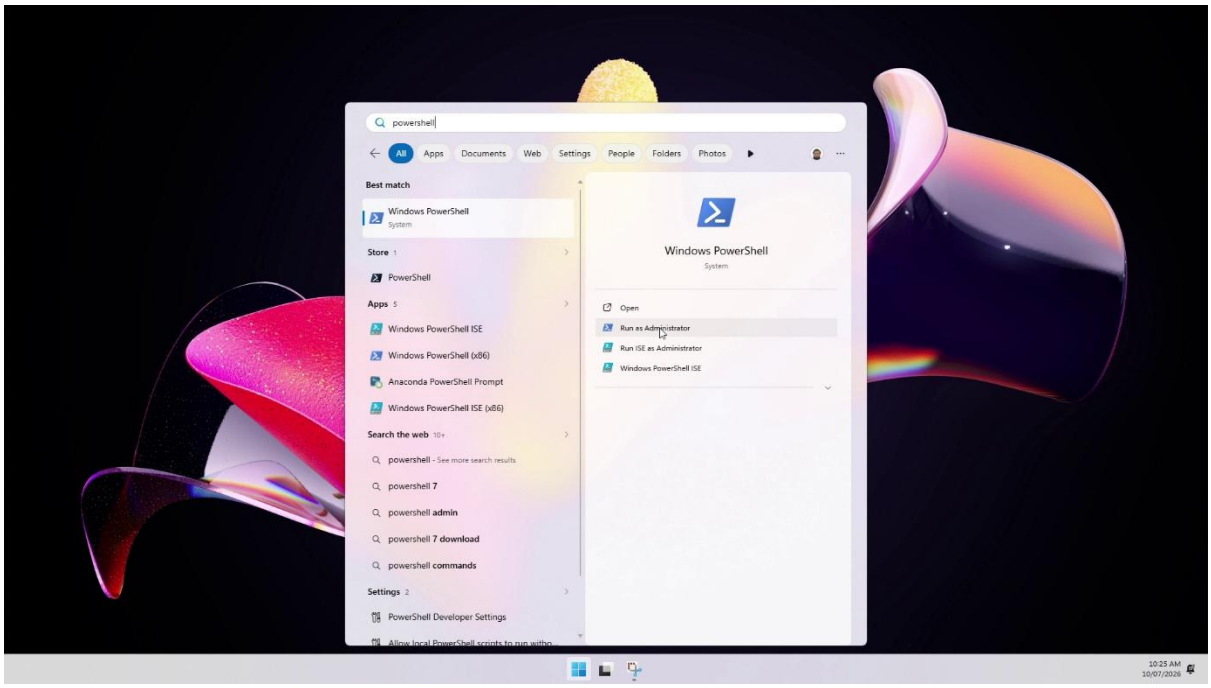




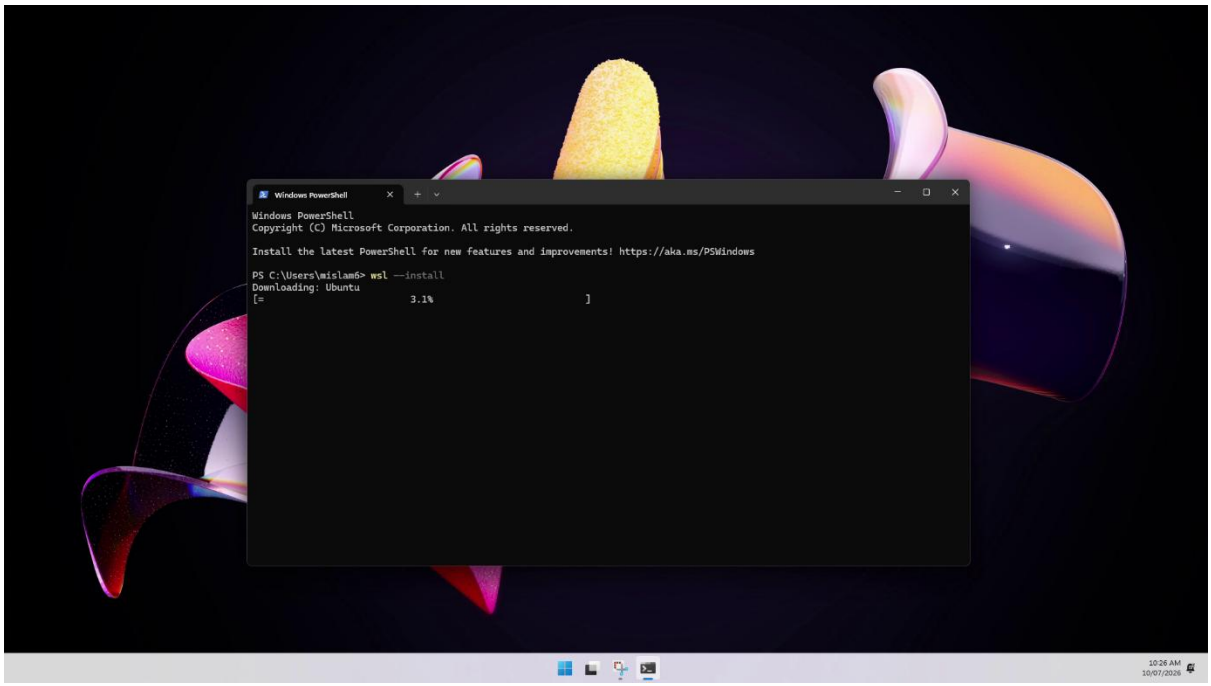
Install Docker on Windows

Step 1: Install WSL 2

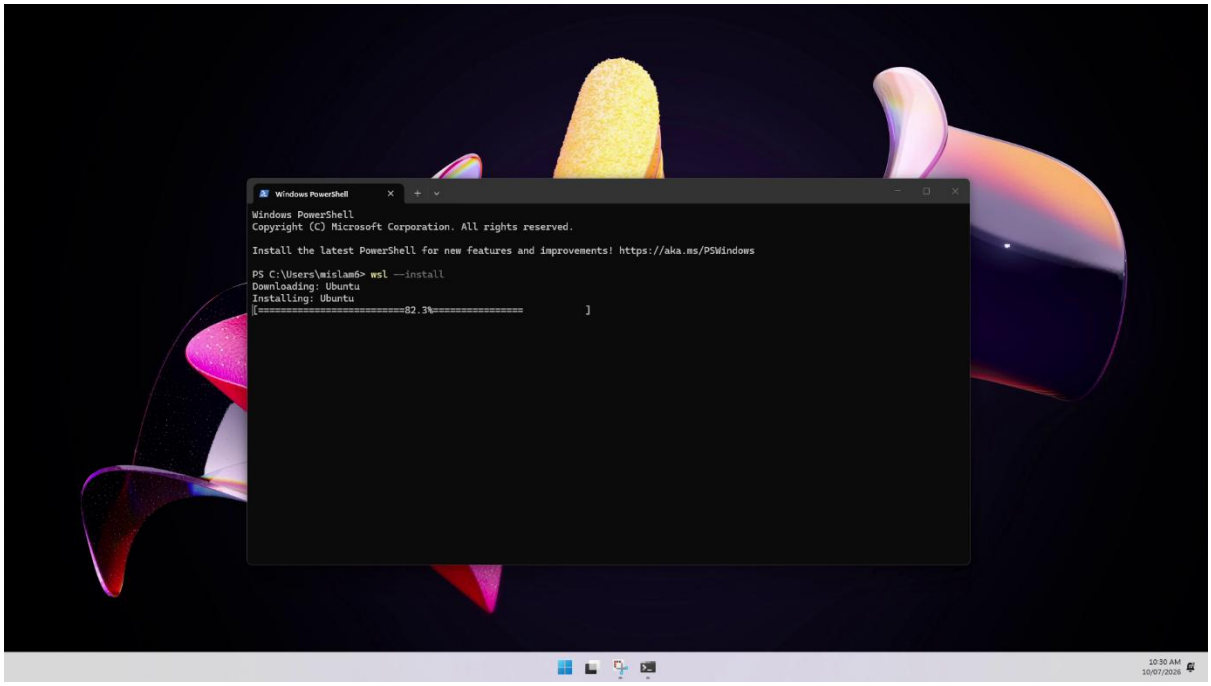
Open PowerShell as Administrator:



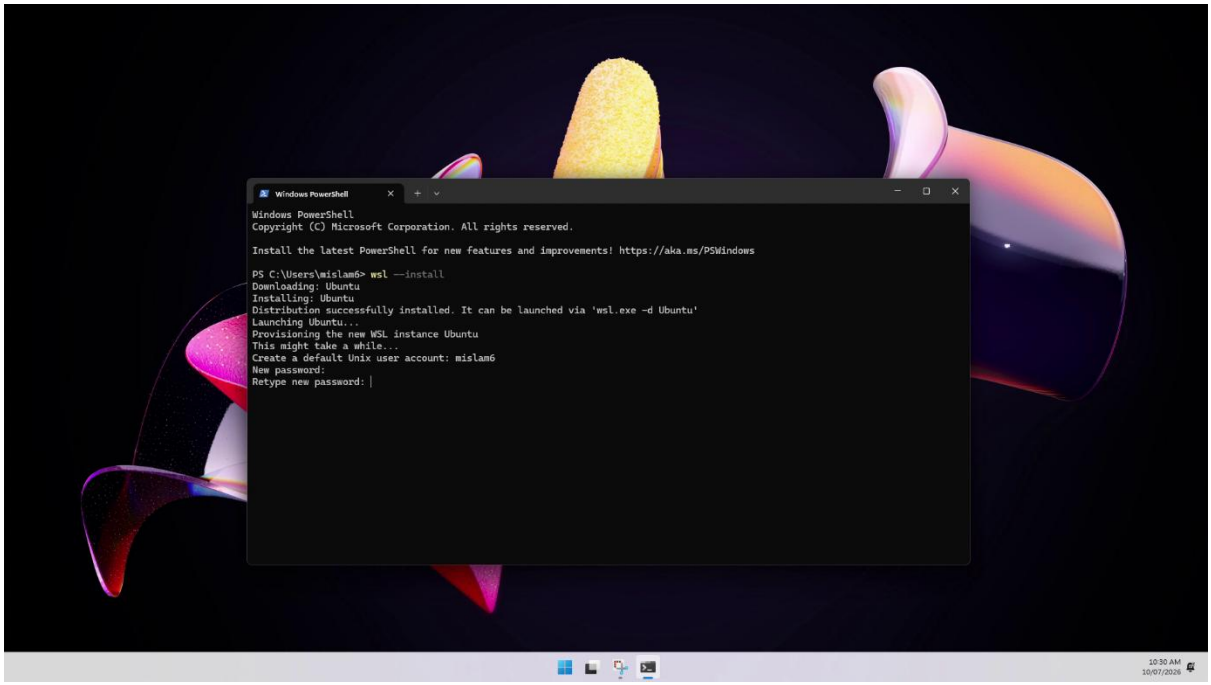
In the “PowerShell” type “wsl --install”



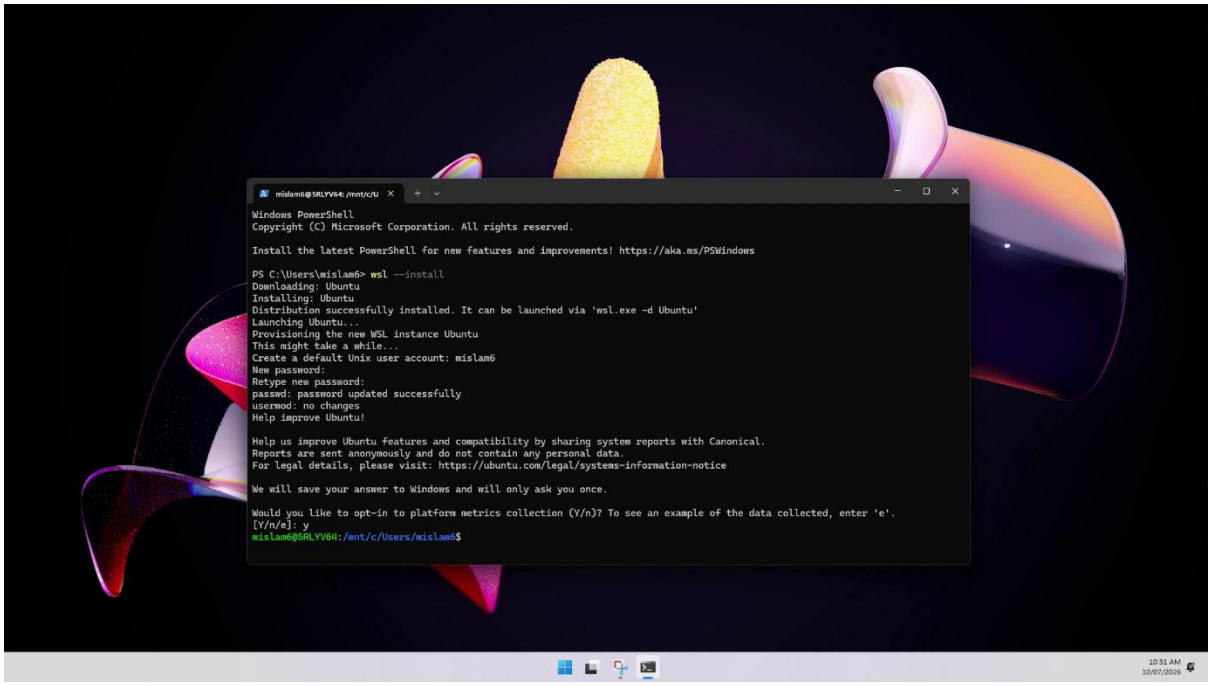
It will start downloading “Ubuntu” Linux on the device.



After the download it will start installing “Ubuntu”



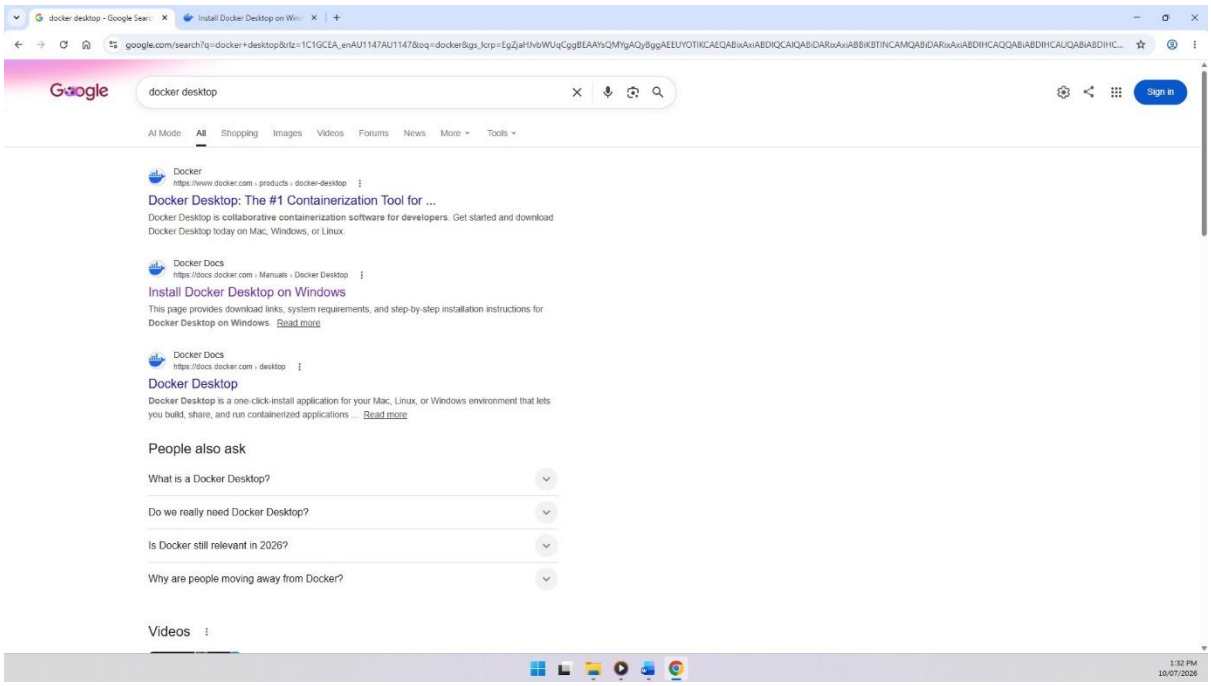
Insert the “User account” name and enter “Password.”



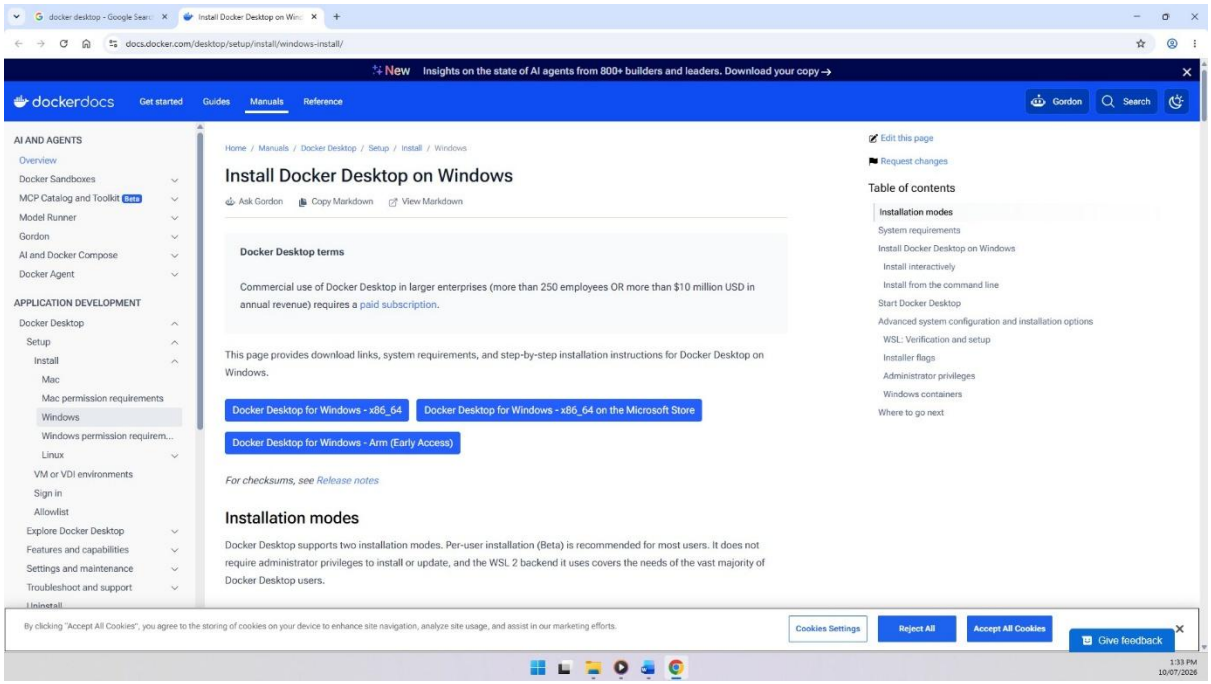
Finally, the “Ubuntu” command line interface will be available through the “PowerShell.”

To quit from the Ubuntu CLI, use the exit command.

Step 2: Download Docker Desktop



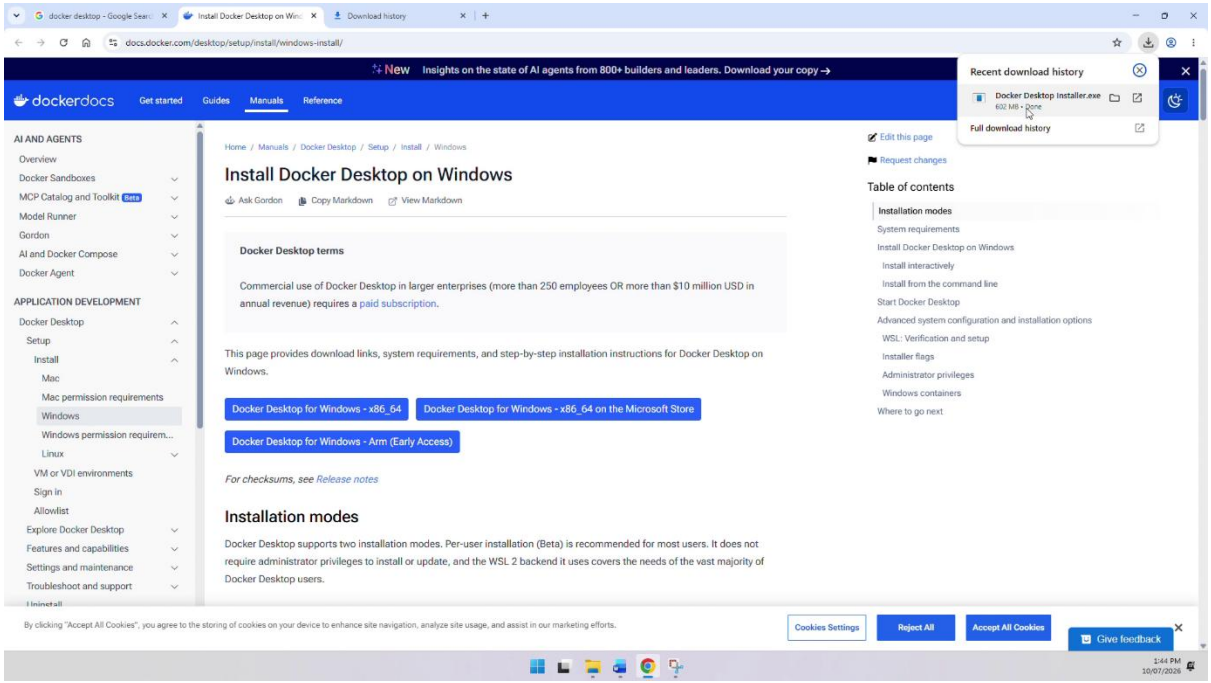
Google “docker desktop” and click “Install Docker Desktop on Windows.”



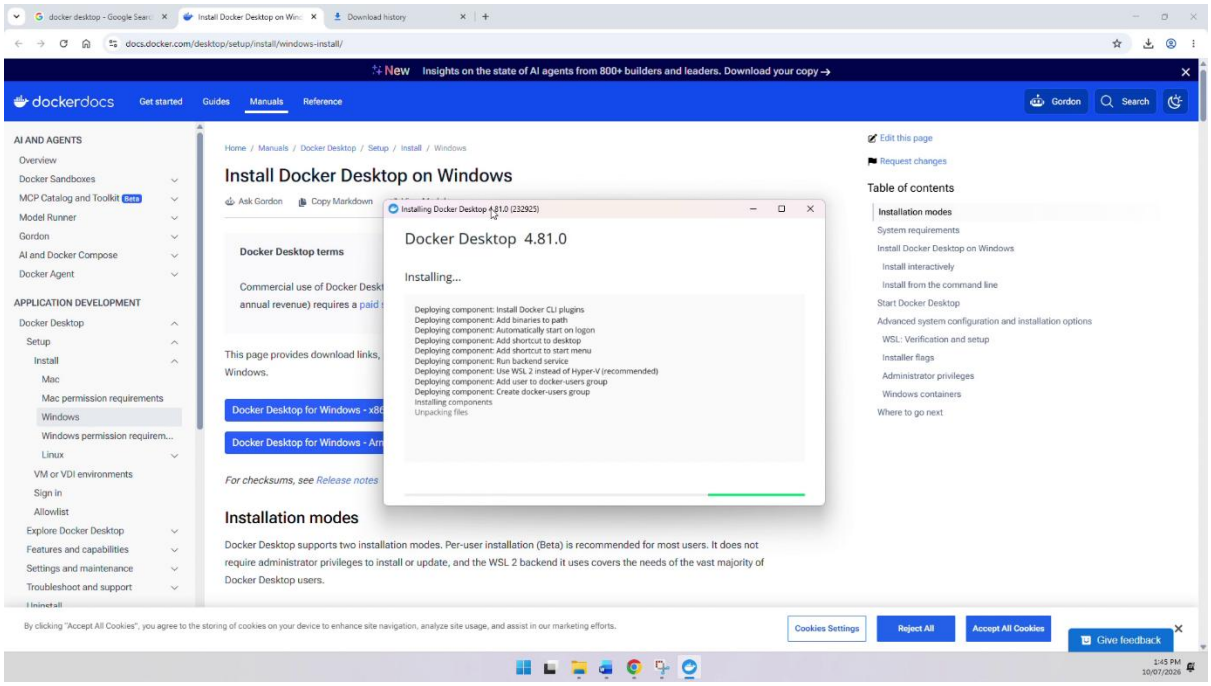
Alternatively, you can go to: https://docs.docker.com/desktop/setup/install/windows-install/

Click “Docker Desktop for Windows - x86_64” to download the “Docker desktop installer.exe”

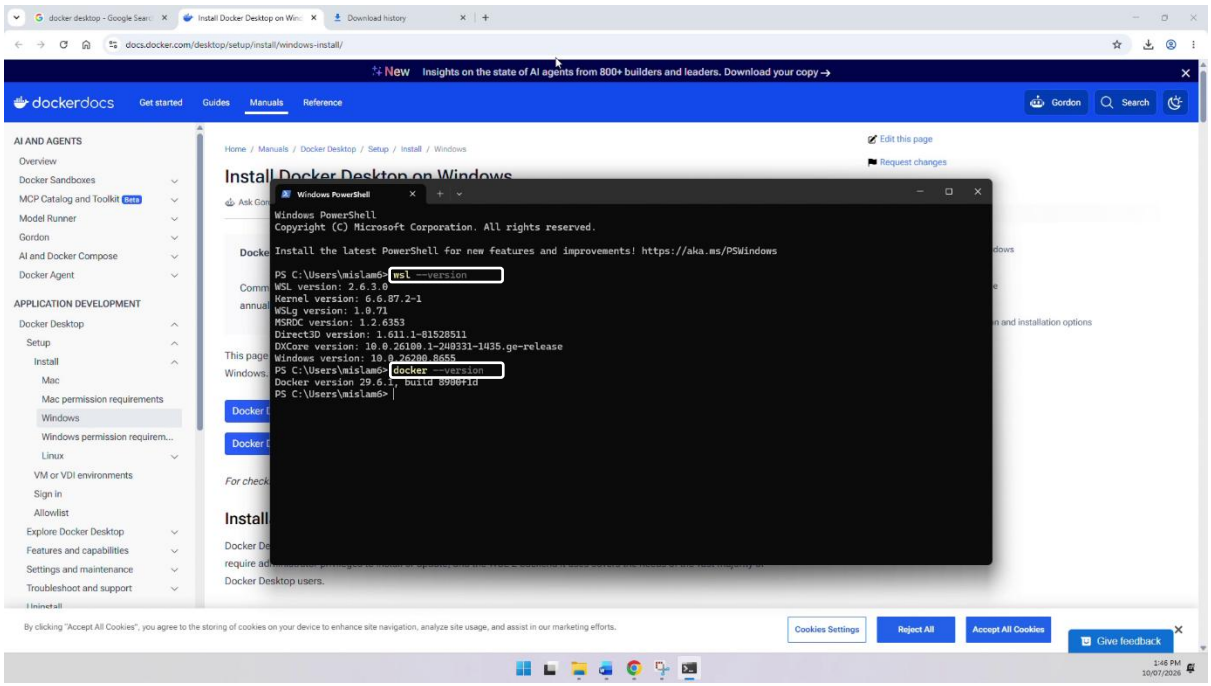
Step 3: Install Docker



Double-click the installer “Docker desktop installer.exe.”



If the setup asks for “Use WSL 2 instead of Hyper-V (recommended)” select this option. Newer version may not ask for it if it detects the WSL on the system.



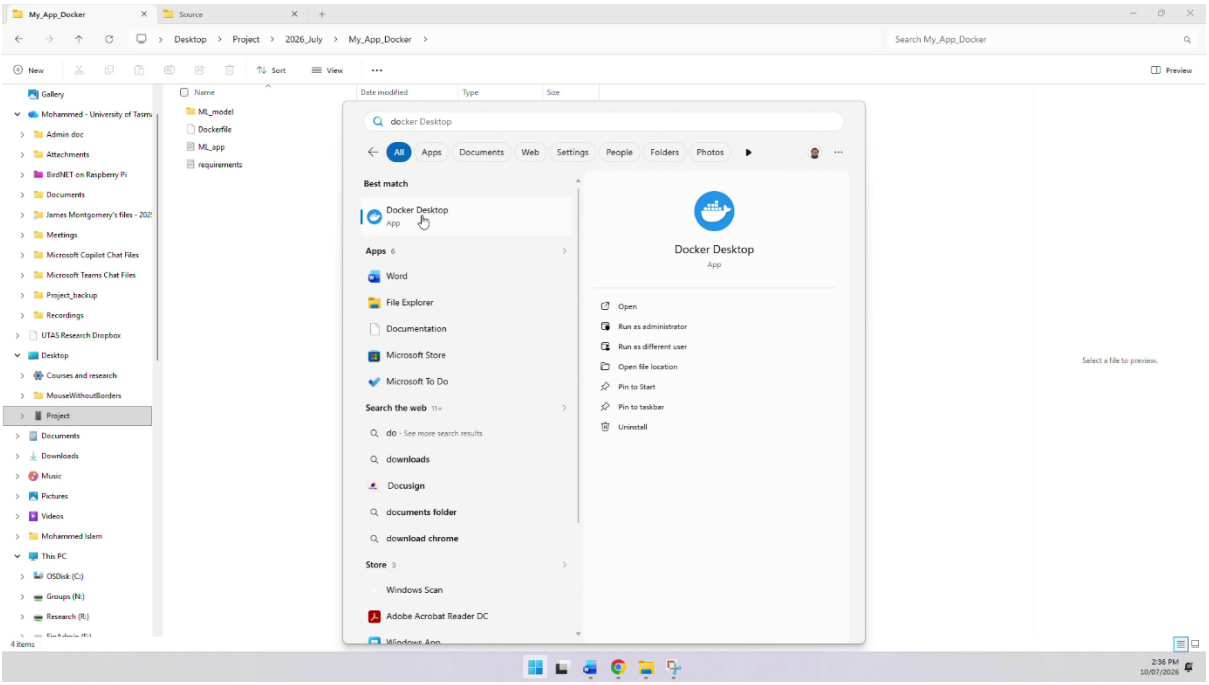
Test the successful installation of WSL and docker by typing:

- wsl --version and
- docker -- version

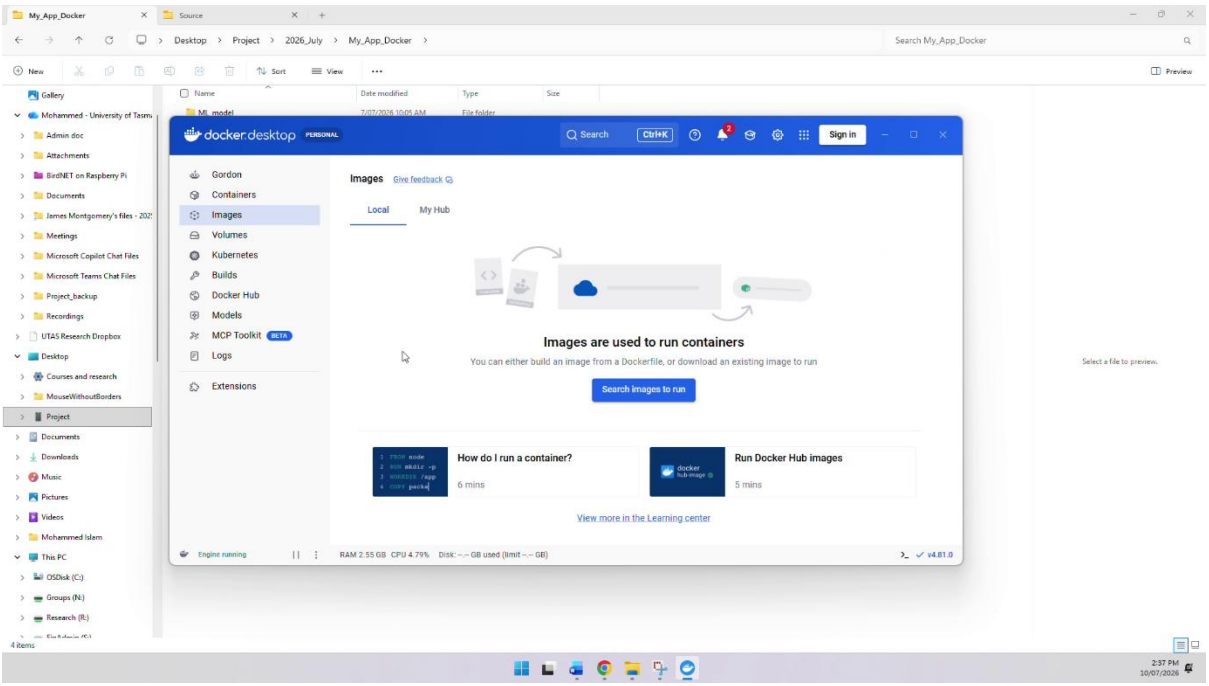
Running these commands will show the

Use Docker to Run the Application

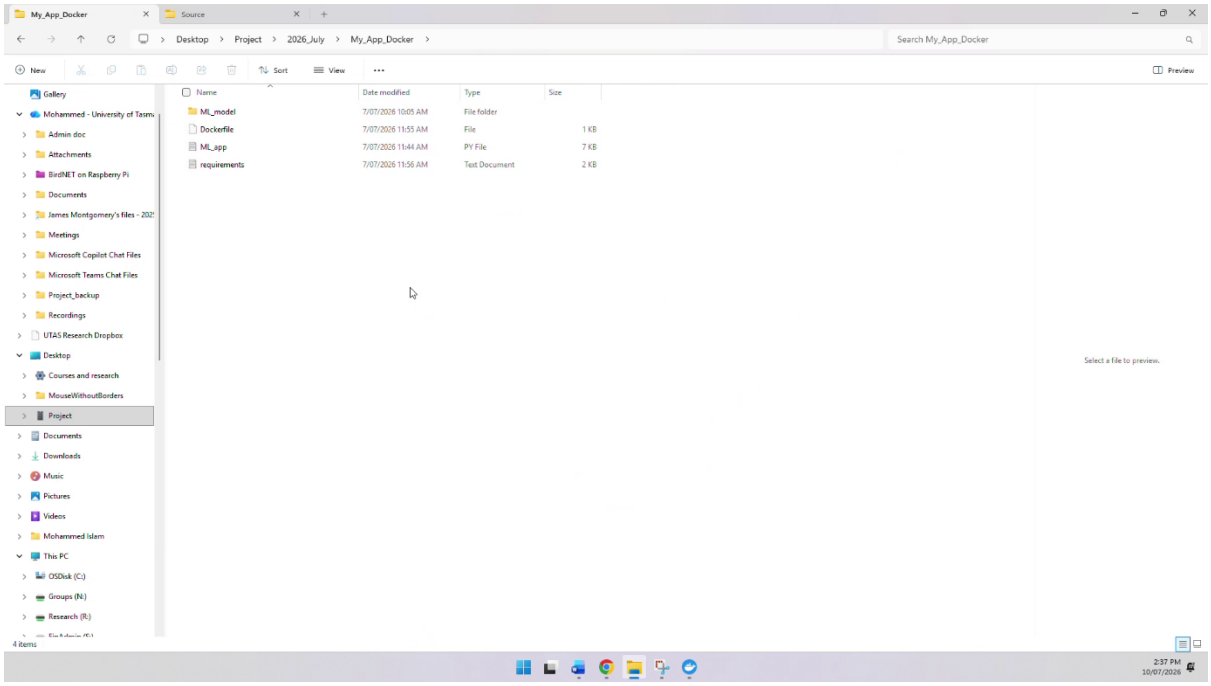
Step 4: Use the application.



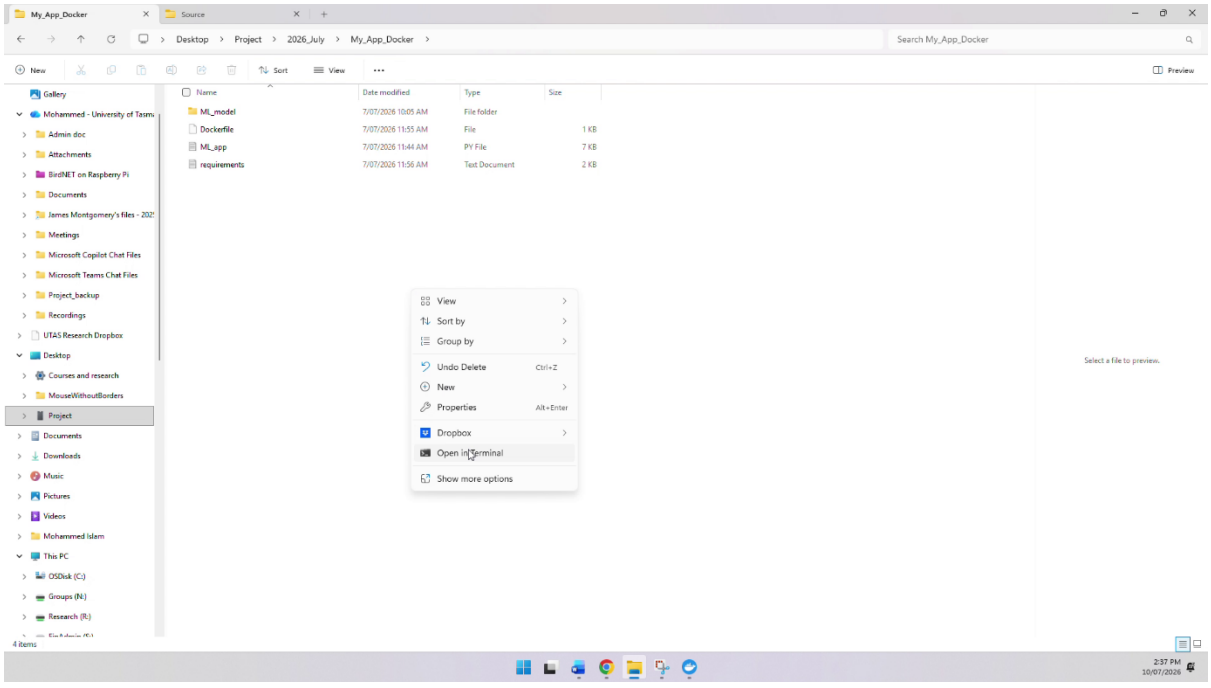
Start Docker from the start menu.



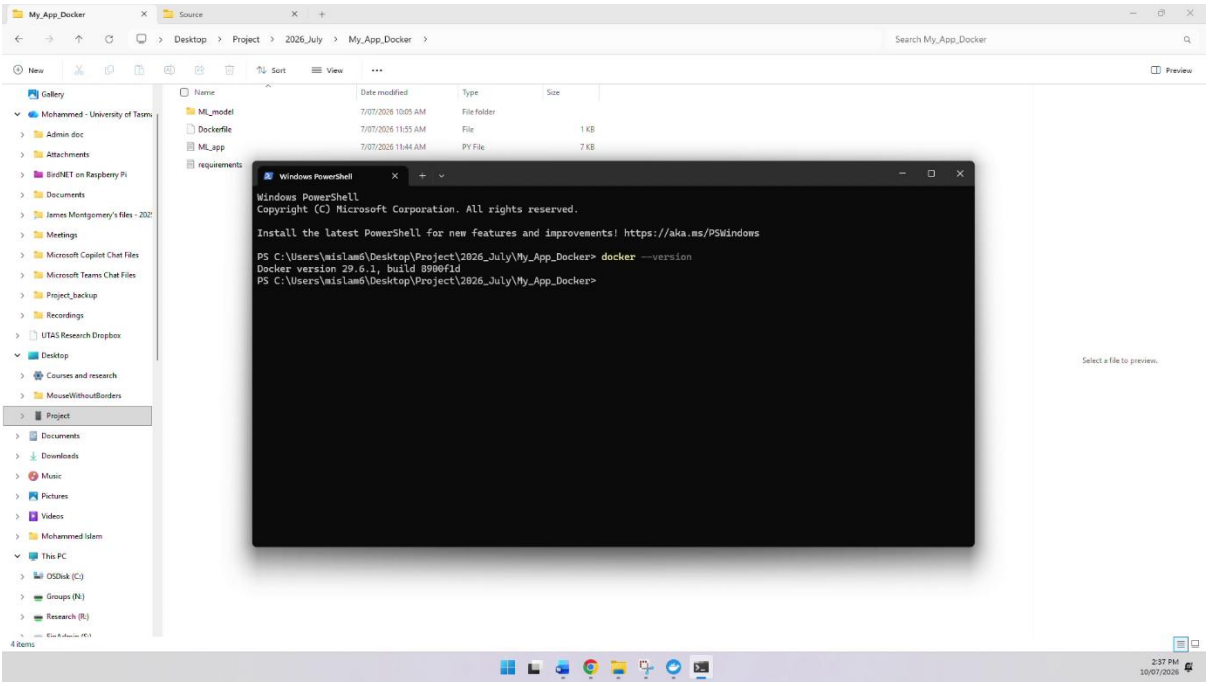
After starting the application should look like this.



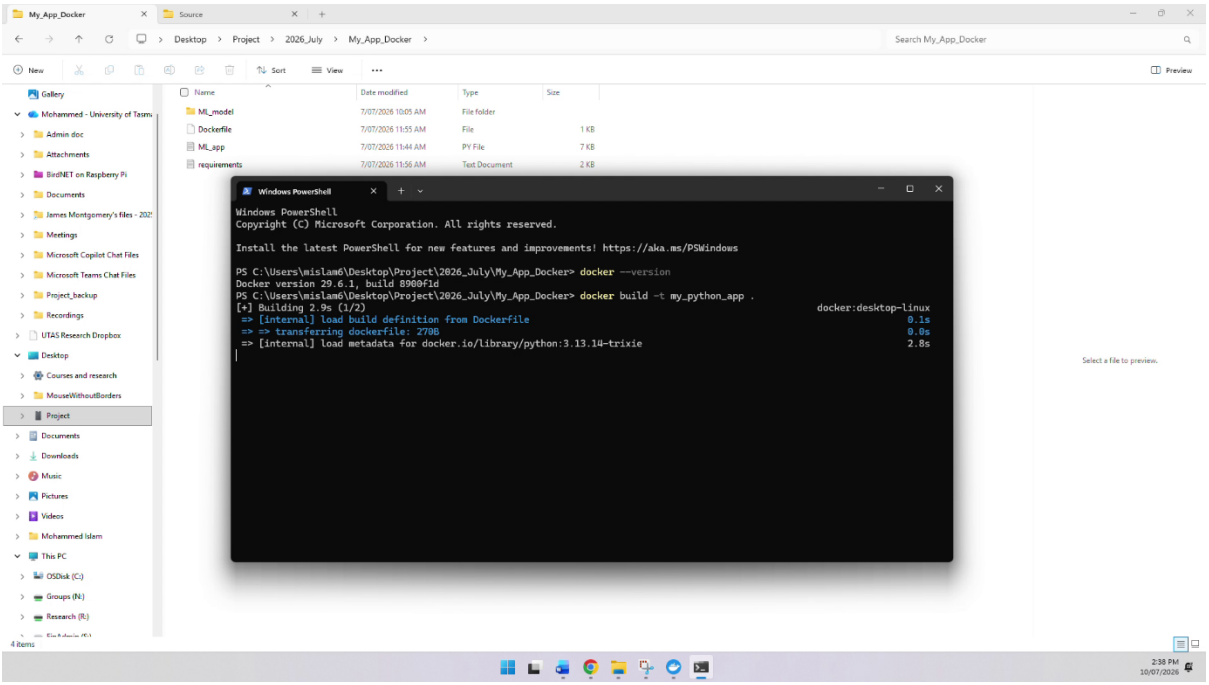
Open the supplied application folder.



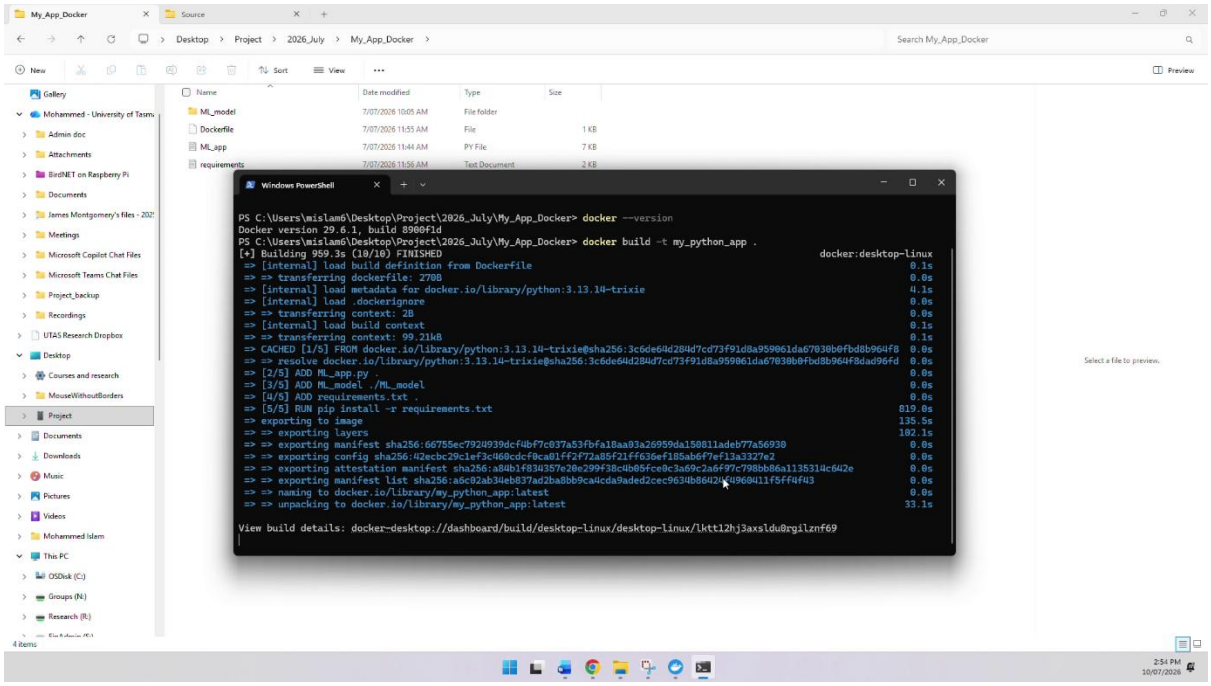
Right click and select “Open in Terminal.”



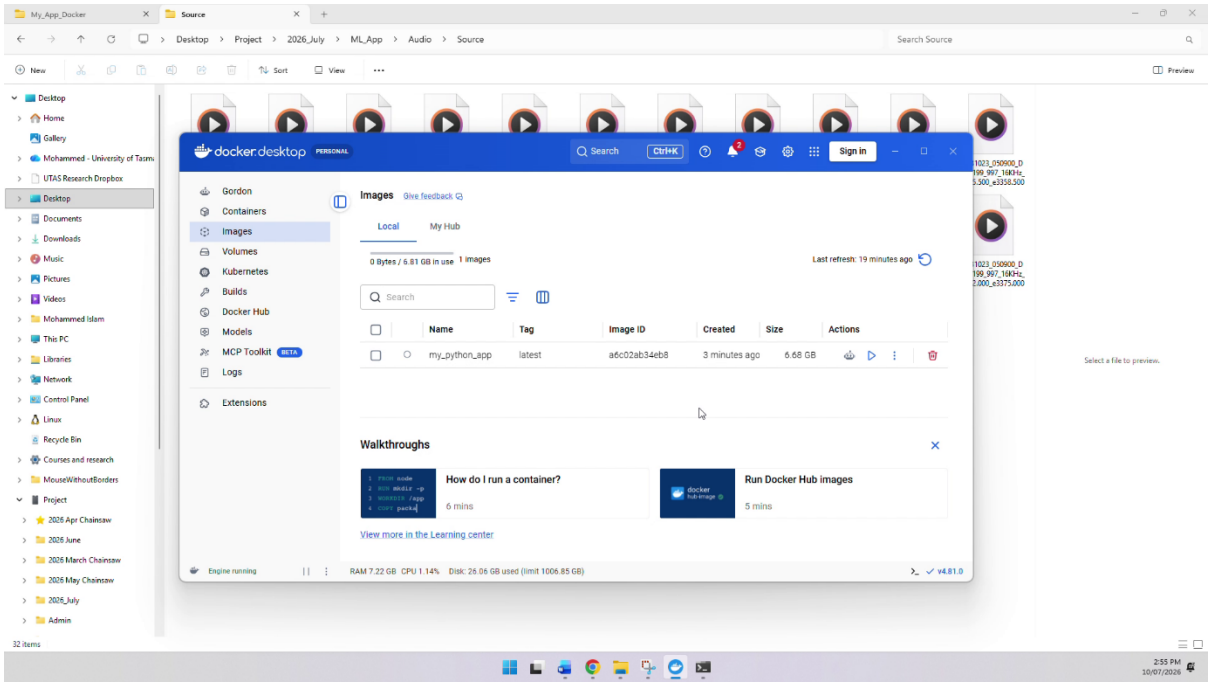
Write “docker --version” in the terminal to make sure docker is available.



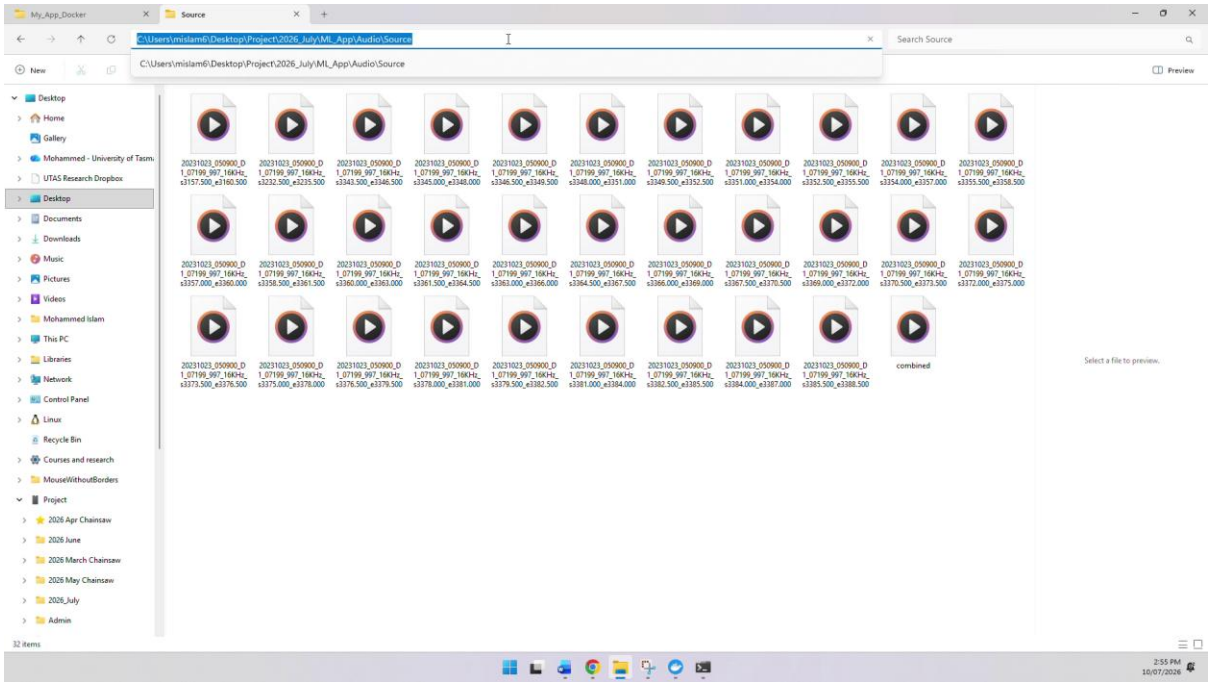
Write “docker build -t my_python_app .” in the terminal to start building the Docker image.



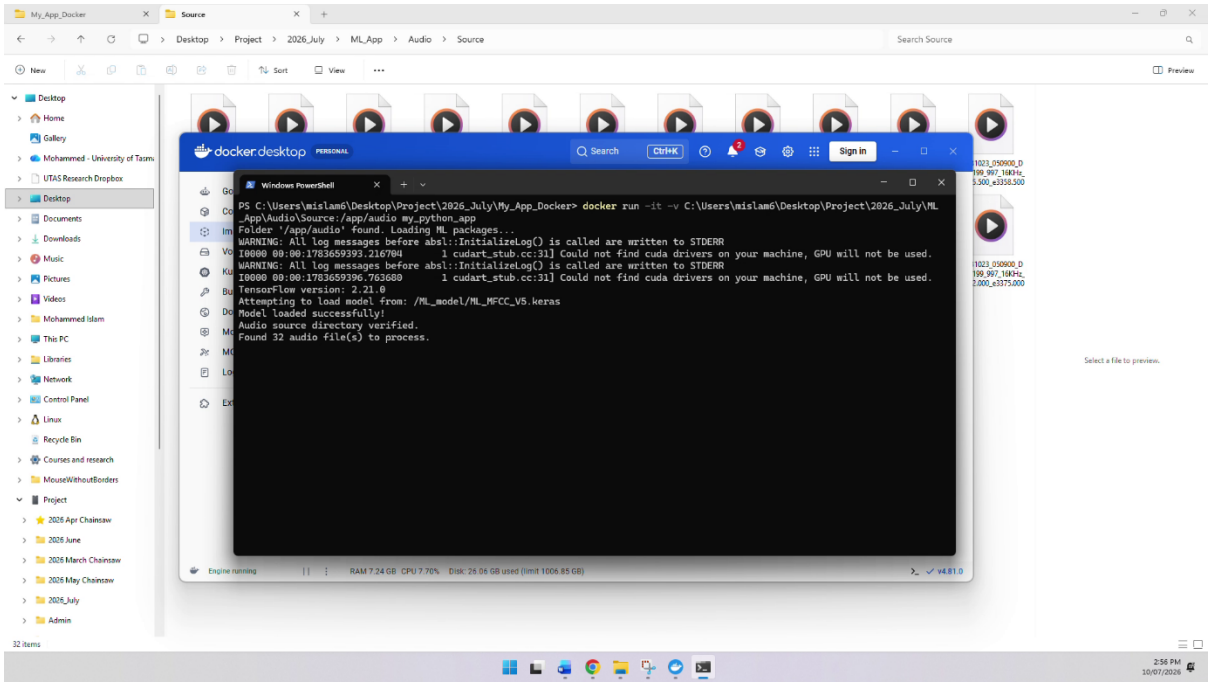
It will take some time to build the image.



Now check the image tag on the Docker desktop application. You will find the built image on the list.

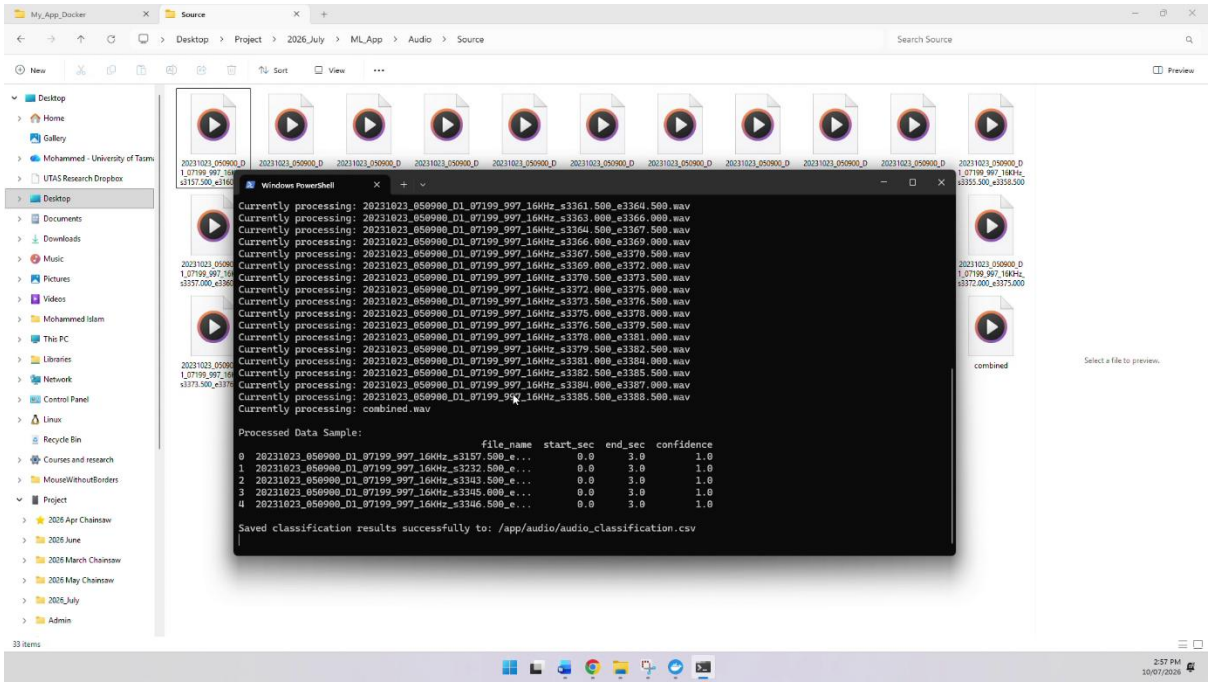


Note the path of the folder that contains the audio files.

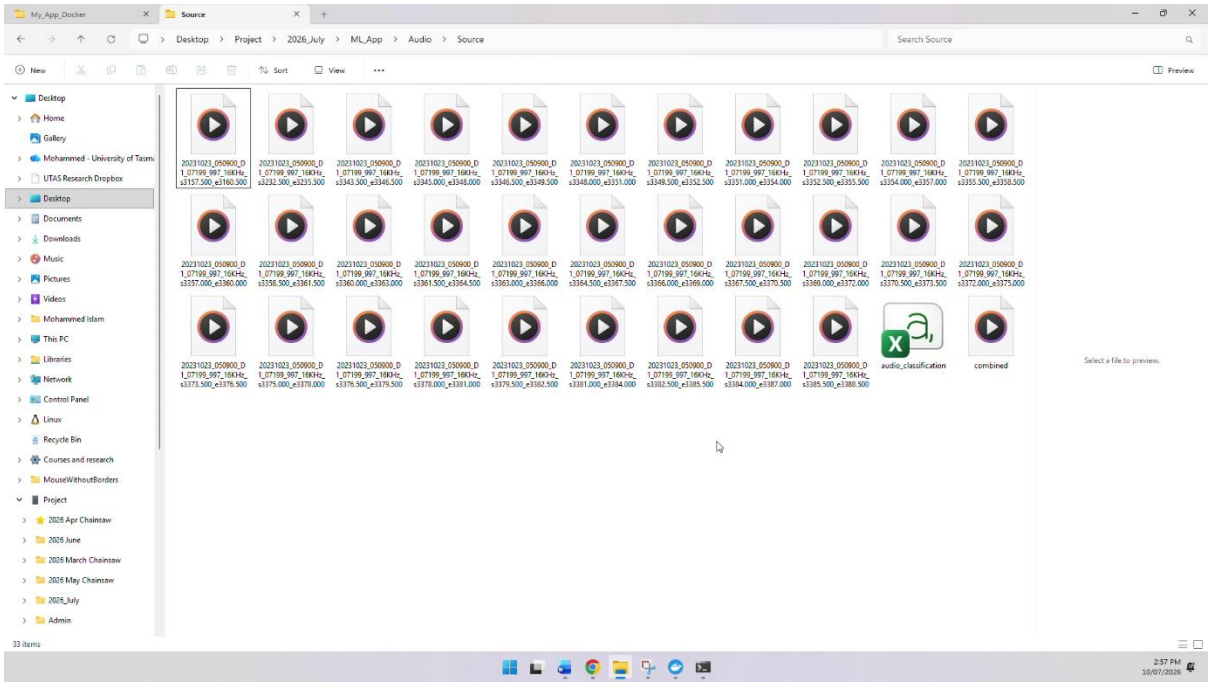


Now run the command “**docker run -it -v** C:\Users\mislam6\Desktop\Project\2026_July\ML_App\Audio\Source**:/app/audio python_app**”

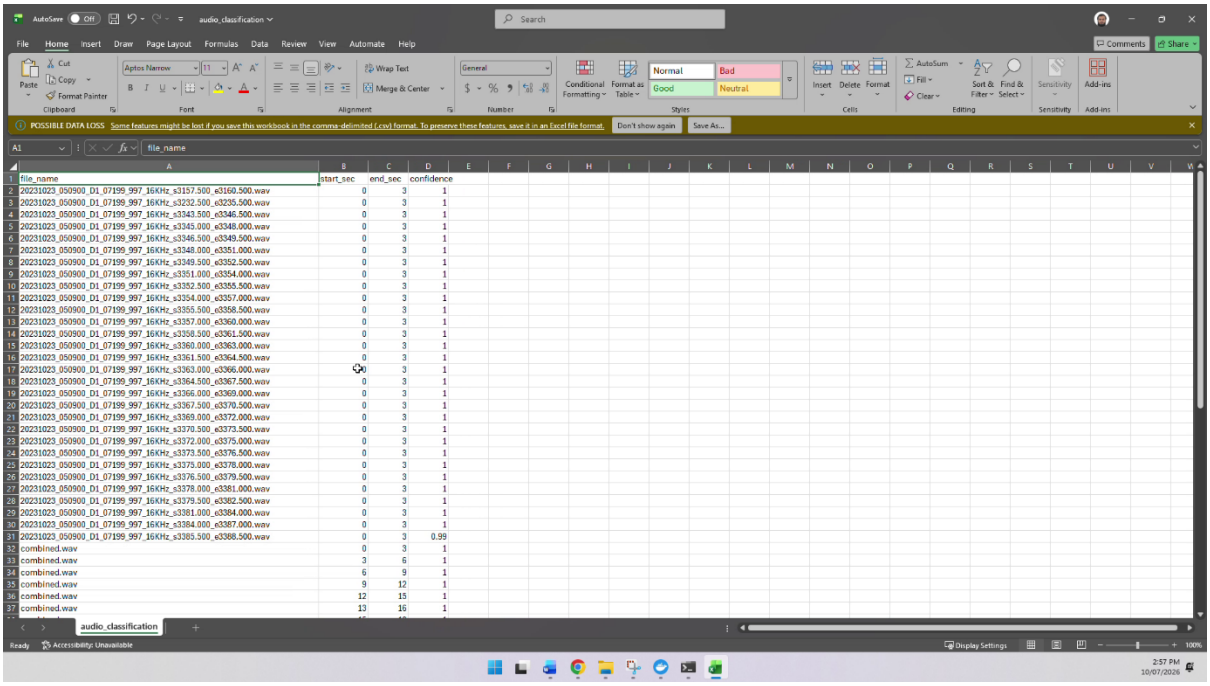
Here, change the audio folder path instead of:
“C:\Users\mislam6\Desktop\Project\2026_July\ML_App\Audio\Source”



It will show the progress of the audio processing.



After the audio processing, there will be a CSV file named “audio_classification.csv” that will have all the classification information.



file_name	start_sec	end_sec	confidence
20231023_050900_D1_07199_997_16KHz_s3157.500_e3160.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3232.500_e3235.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3343.500_e3346.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3345.000_e3348.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3346.500_e3349.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3348.000_e3351.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3349.500_e3352.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3351.000_e3354.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3352.500_e3355.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3354.000_e3357.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3355.500_e3358.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3357.000_e3360.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3358.500_e3361.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3360.000_e3363.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3361.500_e3364.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3363.000_e3366.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3364.500_e3367.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3366.000_e3369.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3367.500_e3370.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3369.000_e3372.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3370.500_e3373.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3372.000_e3375.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3373.500_e3376.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3375.000_e3378.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3376.500_e3379.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3378.000_e3381.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3379.500_e3382.500.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3381.000_e3384.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3384.000_e3387.000.wav	0	3	1
20231023_050900_D1_07199_997_16KHz_s3385.500_e3388.500.wav	0	3	0.99
combined.wav	0	3	1
combined.wav	3	6	1
combined.wav	6	9	1
combined.wav	9	12	1
combined.wav	12	15	1
combined.wav	13	16	1

You can see the file name, start and end time and the confidence ML model thinks is it chainsaw noise.
